

Tree Diagrams and Combined Events

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1

[1 mark]

A fair coin is flipped twice.

How many possible outcomes are there?

1 Each flip has 2 outcomes, and the two flips are independent.

2 $2 \times 2 = 4$ (HH, HT, TH, TT)

Answer: 4

2

[2 marks]

A fair coin is flipped and a fair four-sided dice is rolled.

(a) How many possible outcomes are there?

(b) Write down the probability of getting a head and a 4.

1 (a) 2 outcomes for the coin $\times$ 4 for the dice = 8 outcomes.

2 (b) Only one of the 8 equally likely outcomes is (head, 4).

3 So the probability is $\frac{1}{8}$.

Answer: (a) 8 (b) $\frac{1}{8}$

3

[2 marks]

A and B are independent events with $P(A) = 0.5$ and $P(B) = 0.2$

Work out $P(A \text{ and } B)$.

1 For independent events, multiply along the branches.

2 $0.5 \times 0.2 = 0.1$

Answer: 0.1

4

[2 marks]

A fair coin is flipped twice.

Work out the probability of getting two heads.

1 $P(\text{head}) = \frac{1}{2}$ each time, and the flips are independent.

2 $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

Answer: $\frac{1}{4}$

5

[3 marks]

A bag contains 3 red and 7 blue counters. A counter is taken, its colour recorded, and it is **replaced**. A second counter is then taken.

(a) Work out the probability that both counters are red.

(b) Work out the probability that both counters are blue.

1 Because the counter is replaced, $P(\text{red}) = 0.3$ and $P(\text{blue}) = 0.7$ on both draws.

2 (a) $0.3 \times 0.3 = 0.09$

3 (b) $0.7 \times 0.7 = 0.49$

Answer: (a) 0.09 (b) 0.49

6

[3 marks]

Using the same bag and method (3 red, 7 blue, with replacement, two draws), work out the probability of getting at least one red counter.

1 'At least one red' is the opposite of 'no reds at all'.

2 $P(\text{no reds}) = P(\text{both blue}) = 0.7 \times 0.7 = 0.49$

3 $P(\text{at least one red}) = 1 - 0.49$

4 = 0.51

Answer: 0.51

7

[3 marks]

A spinner has probability 0.3 of landing on 'win'. It is spun twice.

Work out the probability of winning exactly once.

1 There are two ways: win then lose, or lose then win. Both count.

2 Win then lose: $0.3 \times 0.7 = 0.21$

3 Lose then win: $0.7 \times 0.3 = 0.21$

4 Add the two routes: $0.21 + 0.21 = 0.42$

Answer: 0.42

8

[3 marks]

Two fair six-sided dice are rolled.

(a) Work out the probability that both scores are even.

(b) Work out the probability of getting at least one 6.

1 (a) $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$ for each dice.

2 $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

3 (b) Use the complement. $P(\text{not a 6}) = \frac{5}{6}$ each time.

4 $P(\text{no 6s}) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

5 $P(\text{at least one 6}) = 1 - \frac{25}{36} = \frac{11}{36}$

Answer: (a) $\frac{1}{4}$ (b) $\frac{11}{36}$

9

[4 marks]

A bag contains 4 red and 6 blue counters. Two counters are taken at random **without** replacement.

(a) Write down the probability that the first counter is red.

(b) Given that the first counter was red, write down the probability that the second is red.

(c) Work out the probability that both counters are red.

1 (a) There are 10 counters and 4 are red: $\frac{4}{10} = \frac{2}{5}$

2 (b) One red counter has gone, so 3 reds remain out of 9 counters: $\frac{3}{9} = \frac{1}{3}$

3 (c) Multiply along the branches: $\frac{4}{10} \times \frac{3}{9} = \frac{12}{90}$

4 Simplify by dividing by 6: $\frac{2}{15}$

Answer: (a) $\frac{2}{5}$ (b) $\frac{1}{3}$ (c) $\frac{2}{15}$

10

[4 marks]

Using the same bag (4 red, 6 blue, two counters taken without replacement), work out the probability that the two counters are of different colours.

1 There are two successful routes: red then blue, or blue then red.

2 Red then blue: $\frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$

3 Blue then red: $\frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$

4 Add them: $\frac{24}{90} + \frac{24}{90} = \frac{48}{90}$

5 Divide top and bottom by 6: $\frac{8}{15}$

Answer: $\frac{8}{15}$ (about 0.533)

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A and B are independent with $P(A) = 0.6$ and $P(B) = 0.25$

Work out $P(A \text{ and not } B)$.

1 $P(\text{not } B) = 1 - 0.25 = 0.75$

2 $0.6 \times 0.75 = 0.45$

Answer: 0.45

2

[2 marks]

A biased coin has $P(\text{head}) = 0.7$. It is flipped three times.

Work out the probability of getting three heads.

1 The flips are independent, so multiply.

2 $0.7^3 = 0.343$

Answer: 0.343

3

[3 marks]

The same biased coin ($P(\text{head}) = 0.7$) is flipped three times.

Work out the probability of getting at least one head.

1 Use the complement: the opposite of 'at least one head' is 'no heads at all'.

2 $P(\text{tail}) = 0.3$, so $P(\text{three tails}) = 0.3^3 = 0.027$

3 $P(\text{at least one head}) = 1 - 0.027 = 0.973$

Answer: 0.973

4

[3 marks]

A bag contains 5 red and 4 green counters. Two are taken without replacement.

Work out the probability that they are the same colour.

1 Both red: $\frac{5}{9} \times \frac{4}{8} = \frac{20}{72}$

2 Both green: $\frac{4}{9} \times \frac{3}{8} = \frac{12}{72}$

3 Add: $\frac{20}{72} + \frac{12}{72} = \frac{32}{72}$

4 Divide by 8: $\frac{4}{9}$

Answer: $\frac{4}{9}$ (about 0.444)

5

[4 marks]

A factory uses two machines. Machine A makes 70% of the items and 3% of them are faulty. Machine B makes the other 30% and 8% of those are faulty.

Work out the probability that a randomly chosen item is faulty.

1 Draw a tree: first branch A (0.7) or B (0.3), second branch faulty or not.

2 A and faulty: $0.7 \times 0.03 = 0.021$

3 B and faulty: $0.3 \times 0.08 = 0.024$

4 These are the only two ways to get a faulty item, so add them.

5 $0.021 + 0.024 = 0.045$

Answer: 0.045

6

[4 marks]

Using the same factory information, an item is found to be faulty.

Work out the probability that it was made by machine B.

1 This is conditional: restrict attention to the faulty items only.

2 $P(\text{faulty}) = 0.045$ from before.

3 $P(\text{B and faulty}) = 0.024$

4 $P(\text{B} \mid \text{faulty}) = \frac{0.024}{0.045}$

5 $= \frac{24}{45} = \frac{8}{15} \approx 0.533$

Answer: $\frac{8}{15}$ (about 0.533)

7

[4 marks]

A bag contains 6 white and 4 black counters. Three are taken at random without replacement.

Work out the probability that all three are white.

1 Each time a white counter is removed, both the number of whites and the total go down by 1.

2 First: $\frac{6}{10}$ · second: $\frac{5}{9}$ · third: $\frac{4}{8}$

3 Multiply: $\frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} = \frac{120}{720}$

4 $= \frac{1}{6}$

Answer: $\frac{1}{6}$ (about 0.167)

8

[4 marks]

A bag contains n blue counters and 4 red counters.

Two counters are taken at random without replacement. The probability that both are blue is $\frac{1}{3}$

(a) Show that $n^2 - 5n - 6 = 0$

(b) Work out the number of blue counters.

1 (a) There are $n + 4$ counters, so $P(\text{first blue}) = \frac{n}{n + 4}$

2 $P(\text{second blue}) = \frac{n - 1}{n + 3}$

3 $\frac{n(n - 1)}{(n + 4)(n + 3)} = \frac{1}{3}$

4 Cross-multiply: $3n(n - 1) = (n + 4)(n + 3)$, so $3n^2 - 3n = n^2 + 7n + 12$

5 $2n^2 - 10n - 12 = 0$, and dividing by 2 gives $n^2 - 5n - 6 = 0$

6 (b) $(n - 6)(n + 1) = 0$, so $n = 6$ or $n = -1$; a count must be positive, so $n = 6$

7 Check: $\frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3} \checkmark$

Answer: (b) 6 blue counters

9

[4 marks]

A spinner has probability 0.4 of landing on red. It is spun three times.

Work out the probability of getting exactly two reds.

- 1 Exactly two reds can happen in three different orders: RRN, RNR and NRR.
- 2 Each of those has probability $0.4 \times 0.4 \times 0.6 = 0.096$
- 3 There are 3 such orders, so multiply.
- 4 $3 \times 0.096 = 0.288$

Answer: 0.288

10

[5 marks]

Bag A contains 3 red and 5 blue counters. Bag B contains 4 red and 2 blue counters.

One counter is taken at random from bag A and placed into bag B.

A counter is then taken at random from bag B.

Work out the probability that the counter taken from bag B is red.

- 1 Bag B always ends up with 7 counters, but how many are red depends on the first counter.
- 2 **Case 1 — red moved (probability $\frac{3}{8}$):** bag B becomes 5 red and 2 blue.
- 3 $P(\text{red from B}) = \frac{5}{7}$, so this route gives $\frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$
- 4 **Case 2 — blue moved (probability $\frac{5}{8}$):** bag B becomes 4 red and 3 blue.
- 5 $P(\text{red from B}) = \frac{4}{7}$, so this route gives $\frac{5}{8} \times \frac{4}{7} = \frac{20}{56}$
- 6 Add the two routes: $\frac{15}{56} + \frac{20}{56} = \frac{35}{56} = \frac{5}{8}$

Answer: $\frac{5}{8}$ (0.625)